

ALAKSHENDRA J R SINGH

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SKILLS

- **Programming & Development:** Python, Java, C++, Flutter, TypeScript, SQL, Rust
- **Machine Learning:** PyTorch, Computer Vision, Fine Tuning, Natural Language Processing, Generative AI libraries and tools.
- **Tools & Platforms:** HPC Clusters, Unix/Linux, Docker, Microsoft Excel, Slurm, Git
- **Soft Skills:** Team Collaboration, Teaching & Mentoring, Research Writing

EDUCATION

M.Sc. in Computer Science | Technical University of Munich **April 2026 - Mar 2028**

- incoming student at Technical University of Munich (Garching Campus) from Summer Semester 2026.

B.Sc. in Computer Science | Saarland University, Germany **Oct 2021 - Mar 2026**

- one of the few english taught CS bachelor program at a public university which accepted only 15 students for 2021 batch.
- Relevant Coursework: Machine Learning, Database Systems, Software Engineering, Data Structures and Algorithms, Systems Architecture, Theoretical Computer Science, Distributed Computing.

B.Sc. in Artificial Intelligence | Deakin University, Australia **Jul 2020 - Jun 2021**

- *Discontinued after 1 year because of financial reasons during Covid-19 pandemic.*
- I was awarded with Deakin India Merit Scholarship 25% on overall tuition fees.
- Relevant Coursework: Mathematics, Object Oriented Programming, Artificial Intelligence

RESEARCH AND PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant | DFKI, Saarbruecken **Feb 2022 - July 2025**

- Advisor: Dr. Fabrizio Nunnari
- Overall break of 8 months during the tenure due to summer holidays, home country visits, or commitment to other jobs.
- Project 1: The goal was to test the hypothesis that whether Deaf signers represent facial expressions other than defined by Ekman. To test this hypothesis, I extracted ~468 facial landmarks of signers from a sign language dataset. Then using several unsupervised clustering techniques we tried to recognise any specific patterns in clusters. However, cluster formation appeared random and no such pattern could be recognised.

- Project 2: In this project our goal was to beat the state-of-the-art in facial expression recognition on AffectNet Dataset. Due to resource limitation we decided to use cost effective CNN models. I worked on the existing pipeline to implement new image preprocessing techniques and different optimizers, such as Sharpness-Aware Minimization, to improve the recognition accuracy. However, we could not outperform the established state-of-the-art performance.
- Project 3: In this project we tested the viability of using either the upper or lower facial region for accurate recognition of Ekman expressions in signers. I designed a fine-tuning pipeline in Python that leveraged large-scale AffectNet Dataset pretraining before adapting models to a smaller Phoenix sign language dataset. We published our findings that the lower face model preserved most of the predictive power of the full-face model in deaf signers:
<https://doi.org/10.1145/3742886.3756731>

Working Student App Developer | ESCRA GmbH, Saarbruecken **Jan 2024 - Jun 2024**

- Developed a multi-platform app using Flutter for a cybersecurity startup.
- I integrated APIs, authentication, and remote access tools.

Java Software Developer Intern | NIIT Lucknow **May 2019 - Jun 2019**

- Completed an internship before commencing undergraduate studies.
- Developed flight reservation software under the guidance of Senior Java Developer.

PUBLICATION

Nunnari, F., Singh, A. and Gebhard, P. (2025) “Color histogram equalization and fine-tuning to improve expression recognition of (partially occluded) faces on sign language datasets,” in Adjunct Proceedings of the 25th ACM International Conference on Intelligent Virtual Agents. 9th Workshop on Sign Language Translation and Avatar Technology, Berlin, Germany: ACM pp. 1–7 Available at: <https://doi.org/10.1145/3742886.3756731>

SELECTED PROJECTS

Laundry Slot Booking system for Student Dorms - <https://velasp.com> **Dec 2022 - Present**

- Using Flutter and Firebase, I developed a web app to digitize laundry slot bookings in student dorms.
- Secured funding from Student Housing Organisation to address GDPR and legal compliance. App is currently live in one dorm.
- Featured by Saarland University on LinkedIn:
<https://www.linkedin.com/feed/update/urn:li:activity:7329146781038960641/>

Web Application for Saarland University - <https://saaruni.de> **Dec 2023 - Jan 2024**

- Using Flutter and Firebase, I developed a web application for Saarland University to improve internationalization by translating menus of Cafeteria and displaying images. .

AI powered quiz app

Dec 2024 - Mar 2025

- Automated quiz generation from lecture content.
- Used LLM APIs, prompt engineering and JSON parsing.
- Currently in talk with a professor to deploy it next semester as pilot project.

CERTIFICATES

- TensorFlow Developer Professional Certificate | Deeplearning.AI on Coursera
- Deep Learning Specialization | Deeplearning.AI on Coursera
- Mathematics for Machine Learning Specialization | Imperial College London on Coursera
- Computer Vision Masterclass | Udemy
- NLP - Natural Language Processing with Python by Jose Portilla | Udemy
- Artificial Intelligence A-Z: Learn How to build and AI | Udemy
- Flutter & Dart - The Complete Guide [2020 Edition] | Udemy
- Microsoft Excel - Data Visualization, Excel Charts & Graphs by Maven Analytics | Udemy

TEACHING EXPERIENCE

Tutor for Elements of Machine Learning, taught by Prof. Dr. Isabel Valera Winter Semester 2024/25

VOLUNTEER ROLES

Elected Student Council Member Saarland Informatics Campus	Feb 2025 - Present
Event Volunteer World Summit AI, Amsterdam	Oct 2024
Ambassador EU-Startups Summit, Malta	Jan 2025 - Feb 2025
Event Volunteer EBAN - European Business Angel Network, Brussels	Oct 2024